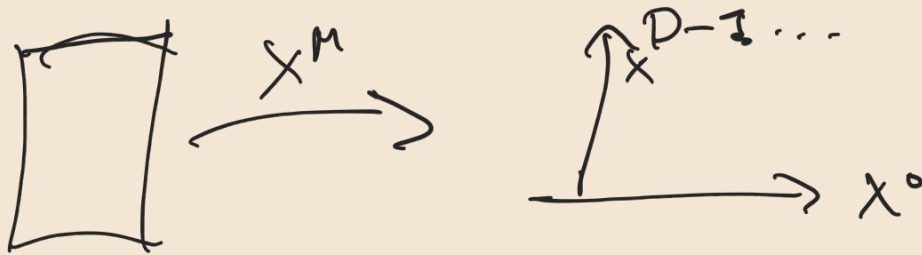


The Dynamics:

The World sheet
has coordinates

$$\xi^\alpha = (\tau, \sigma), \quad \xi^0 = \tau, \quad \xi^1 = \sigma$$



The Worldsheet is Embedded into M^D
via $x^\mu(\xi^\alpha)$.

$$x: \Sigma \rightarrow M^D$$

(i) Open string: $0 \leq \sigma \leq \pi$

(ii) Closed string: $-\pi \leq \sigma \leq \pi$ w/ B.C's
 $x^\mu(\tau, -\pi) = x^\mu(\tau, \pi)$

• Similar to the ^{Point} particle which has
 $S_{p.p.} = -m \int d\tau = -m \int dt \sqrt{-\eta_{\mu\nu} \dot{x}^\mu \dot{x}^\nu}$

We have a generalisation of this
Action for p-branes:

$$S = -T_p \int d\mu_p, \quad T_p: \text{p-brane tension}$$

$$d\mu_p \equiv \sqrt{-\det(G_{\alpha\beta})} d^{p+1}\sigma$$

$d\mu_p: (p+1)\text{-dim volume}$

Where $G_{\alpha\beta}$ is the induced metric

$$G_{\alpha\beta}^I = g_{\mu\nu}(x) \partial_\alpha x^\mu \partial_\beta x^\nu$$

$$\Rightarrow \partial_\alpha x^\mu \equiv \partial x^\mu / \partial \xi^\alpha$$

* Here, $M^D \Rightarrow g_{\mu\nu}(x) = \eta_{\mu\nu}$ thus

$$S_{NG} = -\frac{1}{2\pi\alpha'} \int d^2\xi \sqrt{-\det(\eta_{\mu\nu} \partial_\alpha X^\mu \partial_\beta X^\nu)}$$

- Note: This determinant is over the α, β coordinates and so we would expect

$$\eta_{\mu\nu} \partial_\alpha X^\mu \partial_\beta X^\nu = \eta_{\mu\nu} \begin{pmatrix} \partial_0 X^\mu \partial_0 X^\nu & \partial_0 X^\mu \partial_1 X^\nu \\ \partial_1 X^\mu \partial_0 X^\nu & \partial_1 X^\mu \partial_1 X^\nu \end{pmatrix}$$

$$\Rightarrow \det(G_{\alpha\beta}) = \partial_0 X^\mu \partial_0 X_\mu \partial_1 X^\nu \partial_1 X_\nu - \partial_0 X^\mu \partial_1 X_\mu \partial_1 X^\nu \partial_0 X_\nu$$

- If we move from $\xi^a \mapsto \xi^a + d\xi^a$ on Σ we have a change in M^D given by $X^\mu(\xi^a) \mapsto X^\mu(\xi^a) + d\xi^a \partial_a X^\mu$

The distance in M^D is given by $\eta_{\mu\nu} (d\xi^a \partial_a X^\mu) (d\xi^b \partial_b X^\nu)$

However Note this is simply the induced metric

$$G_{\alpha\beta} = \eta_{\mu\nu} \partial_\alpha X^\mu \partial_\beta X^\nu$$

$$\text{Since } d\xi^a d\xi^b G_{ab} = g_\Sigma$$

* This is very childish, more precisely $X: \Sigma \rightarrow M^D$

$$X^*: T^* M^D \rightarrow T^* \Sigma$$

$$(X^* \eta) (\tilde{X}, \tilde{Y}) = \eta_{\mu\nu} (X_* \tilde{X}, X_* \tilde{Y})$$

$$\begin{aligned} (X_* \tilde{X})^\mu &= \tilde{X}^\alpha \frac{\partial X^\mu}{\partial \xi^\alpha} \\ &= \tilde{X}^\alpha \frac{\partial}{\partial \xi^\alpha} (f \circ X) = \tilde{X}^\alpha \frac{\partial f}{\partial X^\mu} \frac{\partial X^\mu}{\partial \xi^\alpha} \end{aligned}$$

$\therefore$

$$(X^* \eta) (\tilde{X}, \tilde{Y}) = \tilde{X}^\alpha (X^* \eta)_{\alpha\beta} \tilde{Y}^\beta$$

$$LHS = \mathcal{N}(X_* \Sigma, X_* Y)$$

Where $(X_* \Sigma)(f) = \Sigma(f \circ X)$

$$\Sigma^\alpha \partial_\alpha (f \circ X)$$

$$f: \mathbb{M}^D \rightarrow \mathbb{R}$$

$$X: \Sigma \rightarrow \mathbb{M}^D$$

$$f \circ X: \Sigma \rightarrow \mathbb{R}$$

$$\Sigma^\alpha \frac{\partial x^\mu}{\partial \xi^\alpha} \partial_\mu f = (X_* \Sigma)(f)$$

$$\Rightarrow \mathcal{N}(\Sigma^\alpha \frac{\partial x^\mu}{\partial \xi^\alpha} \partial_\mu, \Sigma^\beta \frac{\partial x^\nu}{\partial \xi^\beta} \partial_\nu)$$

$$= \Sigma^\alpha \left[\frac{\partial x^\mu}{\partial \xi^\alpha} \eta_{\mu\nu} \frac{\partial x^\nu}{\partial \xi^\beta} \right] \Sigma^\beta$$

$\therefore$ We conclude

$$(X^* \mathcal{N})_{\alpha\beta} \equiv G_{\alpha\beta}^I = \frac{\partial x^\mu}{\partial \xi^\alpha} \eta_{\mu\nu} \frac{\partial x^\nu}{\partial \xi^\beta}$$

• Introduce the 2D vielbein e_a^α

$$e_a^\alpha \eta_{ab} e_b^\beta = G_{\alpha\beta}^I$$

a : Lorentz indices on Σ

α : Σ world sheet indices

$$\eta_{ab} = \text{diag}(-1, 1)$$

• On the world sheet we have tangent vectors

$$e_a^\alpha \delta \xi^\alpha$$

for which the area is given by

$$\det(e_a^\alpha \delta \xi^\alpha \delta \xi^\alpha)$$

* Note for a static string, the kinetic term vanishes (i.e. all ∂_τ terms)

and $P=0$ so the Hamiltonian

$$H = P\dot{q} - L \Leftrightarrow H = -L$$

$$\therefore S = T \int d\tau d\sigma H \rightarrow T \int d\sigma H$$

Since S is unitless, we find
 $[T] = [E]/[L] \sim [\text{mass}]/[L]$

Symmetries: (Diffeomorphism)

(i) Reparametrizations: $\xi^\alpha \mapsto \xi'^\alpha(\xi)$

$$x^\mu(\xi') = x^\mu(\xi)$$

Infinitesimally: $\delta \xi^\alpha = \xi'^\alpha - \xi^\alpha$

$$\text{i.e. } \xi'^\alpha = \xi^\alpha + \delta \xi^\alpha$$

$$\therefore x^\mu(\xi^\alpha + \delta \xi^\alpha) = x^\mu(\xi) + \partial_\alpha \xi^\alpha \partial_\alpha x^\mu$$

$$\Rightarrow \delta x^\mu = x^\mu(\xi) - x^\mu(\xi) = -\delta \xi^\alpha \partial_\alpha x^\mu$$

In particular we note that the induced metric is also affected

$$G_{\alpha\beta}^I(\xi) = (\partial \xi'^\gamma / \partial \xi^\alpha) (\partial \xi'^\delta / \partial \xi^\beta) G_{\gamma\delta}^I(\xi')$$

(a) $\xi^\alpha \mapsto \xi'^\alpha(\xi), \quad \delta \xi^\alpha = \xi'^\alpha - \xi^\alpha$

(b) $x^\mu(\xi) = x^\mu(\xi')$
 $\delta x^\mu = -\delta \xi^\alpha \partial_\alpha x^\mu$

(c)

* Rep. invariance of the Nambu-Goto Action.

$$S = T \int d^2 \xi \sqrt{-\det(\partial_\alpha x^\mu \partial_\beta x^\nu \eta_{\mu\nu})}$$

Note $S \rightarrow S'$

$$\frac{\partial x^\mu}{\partial \xi'^\alpha} \frac{\partial x^\nu}{\partial \xi'^\beta} = \frac{\partial \xi^\gamma}{\partial \xi'^\alpha} \frac{\partial \xi^\delta}{\partial \xi'^\beta} \partial_\gamma x^\mu \partial_\delta x^\nu \eta_{\mu\nu}$$

Since $x^\mu(\xi') = x^\mu(\xi)$

Putting this together we find

$$S' = T \int d^2 \xi' \sqrt{-\det \left[\frac{\partial \xi}{\partial \xi'} \right]^2 \det [\partial_\alpha x^\mu \partial_\gamma x^\nu]} \\ \det \left[\frac{\partial \xi}{\partial \xi'} \right] \equiv J$$

The Measure transforms as

$$d^2 \xi' = |\partial \xi' / \partial \xi| d^2 \xi = J^{-1} d^2 \xi$$

$$\therefore S' = T \int d^2 \xi J^{-1} J \sqrt{-\det (\partial_\alpha x^\mu \partial_\gamma x^\nu)}$$

(ii) Rigid Poincaré Transform:

$$x^{\mu'}(\xi) = \Lambda^\mu_\nu x^\nu(\xi) + a^\mu$$

$$\Lambda^\mu_\rho \eta_{\mu\nu} \Lambda^\nu_\tau = \eta_{\rho\tau}$$

$$\text{Note } a_{\mu\nu} = -a_{\nu\mu}$$

$$\delta_\rho \eta_{\alpha\beta} = 0$$

Infinitesimally

$$\Lambda^\mu_\nu = \delta^\mu_\nu + a^\mu_\nu, \quad \text{as such}$$

$$\delta_\rho x^\mu \equiv x^{\mu'}(\xi) - x^\mu = (\delta^\mu_\nu + a^\mu_\nu) x^\nu + a^\mu - x^\mu \\ = a^\mu_\nu x^\nu + a^\mu$$

$$|a^\mu_\nu| \ll 1$$

We find $a_{\mu\nu} = -a_{\nu\mu}$ by requiring that $\delta(\eta_{\mu\nu} x^\mu x^\nu) = 0$

* Since Quantizing $\sqrt{\quad}$ is particularly difficult, we introduce a "verlben" equivalent of e for the point particle: a metric $g_{\alpha\beta}$

Polyakov Action:

$$S = -\frac{1}{4\pi\alpha'} \int d^2\xi \sqrt{-g} g^{\alpha\beta} \partial_\alpha x^\mu \partial_\beta x^\nu \eta_{\mu\nu}$$

$g_{\alpha\beta}$: 2D Metric NATURALLY on Σ .
which is Auxiliary.

- Since $\eta_{\mu\nu}$ is from M^D this Action is D-scalar fields x^μ all coupled to two-dimensional gravity $g_{\alpha\beta}$ on Σ .

* This action, unsurprisingly, contains more sym than N.G. Action

(i) Diffeomorphisms: $x^\mu(\xi') = x^\mu(\xi)$

$$g'_{\alpha\beta}(\xi') = \left(\frac{\partial \xi^\gamma}{\partial \xi'^\alpha} \right) \left(\frac{\partial \xi^\delta}{\partial \xi'^\beta} \right) g_{\gamma\delta}(\xi)$$

(ii) Poincare: (As above)

(iii) Weyl:

$$g'_{\alpha\beta} = e^{\phi(\sigma, \varepsilon)} g_{\alpha\beta}, \quad x'^\mu(\xi) = x^\mu(\xi)$$

E.O.M.: $S = T \int d^2\xi \sqrt{-g} g^{\alpha\beta} \partial_\alpha x^\mu \partial_\beta x^\nu \eta_{\mu\nu}$

Our Lagrangian (density)

$$\mathcal{L} = T \sqrt{-g} g^{\alpha\beta} \partial_\alpha x^\mu \partial_\beta x^\nu \eta_{\mu\nu}$$

$$\therefore 0 = \delta S = \int_a^\pi d\sigma \int_{\tau_1}^{\tau_2} d\tau \left[\frac{\partial \mathcal{L}}{\partial g_{\alpha\beta}} \delta g_{\alpha\beta} + \frac{\partial \mathcal{L}}{\partial \partial_\alpha x^\mu} \delta(\partial_\alpha x^\mu) \right]$$

The second term expands over the α index:

$$\frac{\partial \mathcal{L}}{\partial (\partial_\alpha x^\mu)} \delta(\partial_\alpha x^\mu) = \frac{\partial \mathcal{L}}{\partial \partial_\tau x^\mu} \delta(\partial_\tau x^\mu) + \frac{\partial \mathcal{L}}{\partial \partial_\sigma x^\mu} \delta(\partial_\sigma x^\mu)$$

→ Require that $\delta x^1 = \delta x^2 = \delta g_{\alpha\beta} = 0$

$$0 = \int_a^\pi d\sigma \int_{\tau_1}^{\tau_2} d\tau \left\{ \frac{\partial \mathcal{L}}{\partial g_{\alpha\beta}} \delta g_{\alpha\beta} + \frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} \delta \dot{x}^\mu + \frac{\partial \mathcal{L}}{\partial x^\mu} \delta x^\mu \right\}$$

Integrate by parts so that

$$\begin{aligned} \textcircled{a} \int_a^\pi d\sigma \int_{\tau_1}^{\tau_2} d\tau \frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} \partial_\tau (\delta x^\mu) &= \iint d\tau d\sigma \frac{d}{d\tau} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} \delta x^\mu \right) \\ &= \int_a^\pi d\sigma \left[\frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} \delta x^\mu \right]_{\tau_1}^{\tau_2} - \int_a^\pi d\sigma \int_{\tau_1}^{\tau_2} d\tau \partial_\tau \frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} \delta x^\mu \end{aligned}$$

But $\delta x^\mu(\tau_1) = \delta x^\mu(\tau_2) = 0$

$$\therefore \textcircled{a} = - \int_a^\pi d\sigma \int_{\tau_1}^{\tau_2} d\tau \partial_\tau \frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} \delta x^\mu$$

$$\textcircled{b} = \int_{\tau_1}^{\tau_2} d\tau \left[\frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} \delta x^\mu \right]_{\sigma=a}^{\sigma=\pi} - \int_a^\pi d\sigma \int_{\tau_1}^{\tau_2} d\tau \left(\partial_\sigma \frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} \delta x^\mu \right)$$

$$\therefore 0 = \iint_{\substack{\sigma=a \\ \tau_1}}^{\substack{\sigma=\pi \\ \tau_2}} d\sigma d\tau \left\{ \frac{\partial \mathcal{L}}{\partial g_{\alpha\beta}} \delta g_{\alpha\beta} - \partial_\sigma \left(\frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} \right) \delta x^\mu - \partial_\tau \left(\frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} \right) \delta x^\mu \right\} + \int_{\tau_1}^{\tau_2} d\tau \left[\frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} \delta x^\mu \right]_{\sigma=a}^{\sigma=\pi}$$

Case (i): $\delta x^\mu = 0$, $\delta g_{\alpha\beta}$: arbitrary

$$\Rightarrow \boxed{\partial \mathcal{L} / \partial g_{\alpha\beta} = 0} \quad \textcircled{a}$$

Case (ii): $\delta g_{\alpha\beta} = 0$, δx^μ : arbitrary up to $\delta x^\mu(\pi) = \delta x^\mu(a) = 0$

$$\textcircled{b} \quad \boxed{\partial_\alpha \pi^{\alpha\mu} = 0} \quad \text{where} \quad \boxed{\pi^{\alpha\mu} = \frac{\partial \mathcal{L}}{\partial \dot{x}^\mu}}$$

$$\textcircled{a} \quad \frac{\partial \mathcal{L}}{\partial g_{\alpha\beta}} = 0 = T \sqrt{-g} \partial^\alpha x^\mu \partial^\beta x_\mu + \dots$$

note $\delta \sqrt{-g} = -\frac{1}{2} \frac{1}{\sqrt{-g}} g^{\alpha\beta} \delta g_{\alpha\beta} = -\frac{1}{2} \sqrt{-g} g^{\alpha\beta} \delta g_{\alpha\beta}$

$$\frac{\partial \mathcal{L}}{\partial g^{\alpha\beta}} = \sqrt{-g} \partial^\alpha X^\mu \partial^\beta X_\mu - \frac{1}{2} g^{\alpha\beta} \partial_\gamma X^\mu \partial_\delta X_\mu g^{\gamma\delta}$$

$$\therefore \boxed{0 = \partial_\alpha X^\mu \partial_\beta X_\mu - \frac{1}{2} g_{\alpha\beta} g^{\gamma\delta} \partial_\gamma X^\mu \partial_\delta X_\mu}$$

$$\textcircled{b} \partial_\tau \left(\frac{\partial \mathcal{L}}{\partial \partial_\tau X^\mu} \right) + \partial_\sigma \left(\frac{\partial \mathcal{L}}{\partial \partial_\sigma X^\mu} \right)$$

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \partial_\sigma X^\mu} &= \frac{\partial}{\partial \partial_\sigma X^\mu} \left[-T \sqrt{-g} g^{\alpha\beta} \eta^{\mu\nu} \partial_\alpha X_\mu \partial_\beta X_\nu \right] \\ &= 2 \sqrt{-g} T g^{\alpha\beta} \eta^{\mu\nu} \delta_\alpha^\sigma \delta_\mu^\lambda \\ &= (2 \sqrt{-g} T) (g^{\sigma\beta} \eta^{\lambda\nu} \partial_\beta X_\nu) \end{aligned}$$

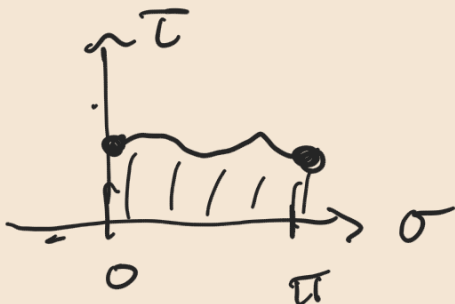
$$\therefore \boxed{\partial_\alpha \pi^{\alpha\mu} = \partial_\alpha (\sqrt{-g} g^{\alpha\beta} \partial_\beta X^\mu) = 0}$$

* For the closed string: $X^\mu(\tau, \sigma) = X^\mu(\tau, \sigma + 2\pi)$
 so $\delta X^\mu(-\pi) = \delta X^\mu(\pi)$ and so the Boundary term vanishes

* For the open string we take

$$\partial_\alpha (\sqrt{-g} g^{\alpha\beta} \partial_\beta X^\mu) = 0$$

to hold at the Boundary $\sigma=0, \sigma=\pi$



→ There is no σ restriction on δX^μ this implies $\delta S = 0 \iff$

$$\pi^{\alpha\mu} \equiv \frac{\partial \mathcal{L}}{\partial \partial_\alpha X^\mu} = 0 \quad \text{at } \sigma=0 \text{ \& } \sigma=\pi$$

We have, from above

$$\tau^{1m} = \sqrt{-g} g^{1B} \partial_B x^m = 0$$

$$\therefore \boxed{\sqrt{-g} g^{1B} \partial_B x^m = 0} \text{ at } \sigma=0, \sigma=\pi$$

that is,

$$\sqrt{-g} g^{1B} \partial_B x^m \big|_{\sigma=0, \pi} = 0$$

This is really two separate equations:

→ There is no relationship between the two ends of the string.

* Now, from the $g_{\alpha\beta}$ E.O.M we know

$$\frac{1}{2} g_{\alpha\beta} f^{-1} = \partial_\alpha x^m \partial_\beta x_m \text{ or}$$

$$\boxed{g_{\alpha\beta} = 2f \partial_\alpha x^m \partial_\beta x_m}, \quad \boxed{f^{-1} \equiv g^{\alpha\beta} \partial_\alpha x^m \partial_\beta x_m}$$

$$\begin{aligned} \text{Thus, } g &= \det(g_{\alpha\beta}) = (2f)^2 \det(\partial_\alpha x^m \partial_\beta x^N \eta_{mn}) \\ &= 4f^2 \det(G_{\alpha\beta}) \end{aligned}$$

→ Side Note: $g_{\alpha\beta}$ is Conformally Eq. to the induced metric $G_{\alpha\beta}$

$$\sqrt{-g} = 2f \sqrt{-\det(G_{\alpha\beta})}$$

$$\text{or } \sqrt{-g} f^{-1} = 2 \sqrt{-\det(G_{\alpha\beta})}$$

$$\boxed{\sqrt{-g} g^{\alpha\beta} \partial_\alpha x^m \partial_\beta x^N = 2 \sqrt{-\det(G_{\alpha\beta})}}$$

↳ This equivalence allows us to understand the Equivalence of N.G. & P.S. actions at the Classical level.

Gauge Choices:

Riemann surface: 2D surface w/ a Euclidean

metric

ii) On any Riemannian surface one can always locally find a f.t. $\phi(\xi)$ s.t.
 $g_{\alpha\beta}(\xi) = \eta_{\alpha\beta} e^{\phi(\xi)}$

→ Physics: Counting D.O.F. $g_{\alpha\beta}$ is symmetric and $g^{\alpha\beta}g_{\alpha\beta} = 2$

$$\therefore g_{\alpha\beta} = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \Rightarrow \alpha + \delta = 2$$

so $g_{\alpha\beta}$ has 3 D.O.F. However

$g^c_{\alpha\beta} = e^{\phi(\xi)} \eta_{\alpha\beta}$ has 1 D.O.F due to $\phi(\xi)$

* Note: Cannot make a Gauge choice Before we have taken the variation of S .

→ Impose gauge choices on the E.O.M

$$\Leftrightarrow \partial_\alpha (\sqrt{-g} g^{\alpha\beta} \partial_\beta x^\mu) = 0$$

In the Conformal Gauge

$$g^c_{\alpha\beta} = e^{\phi(\xi)} \eta_{\alpha\beta}$$

$$\partial_\alpha (e^{\phi} e^{-\phi} \eta^{\alpha\beta} \partial_\beta x^\mu) = \partial_\alpha (\eta^{\alpha\beta} \partial_\beta x^\mu) = 0$$

massless

• This leads to the wave-equation

$$\boxed{\partial^2 x^\mu = 0}$$

• The Second E.O.M: $\partial_\alpha x^\mu \partial_\beta x_\mu - \frac{1}{2} g_{\alpha\beta} g^{\lambda\delta} \partial_\delta x^\mu \partial_\lambda x_\mu = 0$

Becomes

$$\partial_\alpha x^\mu \partial_\beta x_\mu - \frac{1}{2} e^{\phi} \eta_{\alpha\beta} e^{-\phi} \eta^{\delta\lambda} \partial_\delta x^\mu \partial_\lambda x_\mu = 0$$

$$\boxed{T_{\alpha\beta} \equiv \partial_\alpha x^\mu \partial_\beta x_\mu - \frac{1}{2} \eta_{\alpha\beta} \partial_\alpha x^\mu \partial^\alpha x_\mu = 0}$$

* We note that $T_{\alpha\beta}$ is also traceless:

$$\eta^{\alpha\beta} T_{\alpha\beta} = 0 \quad (\text{trivially})$$

$$T_{00} = T_{11} = \frac{1}{2} (\dot{x}^\mu \dot{x}^\nu + x^{\mu'} x^{\nu'}) \eta_{\mu\nu}$$

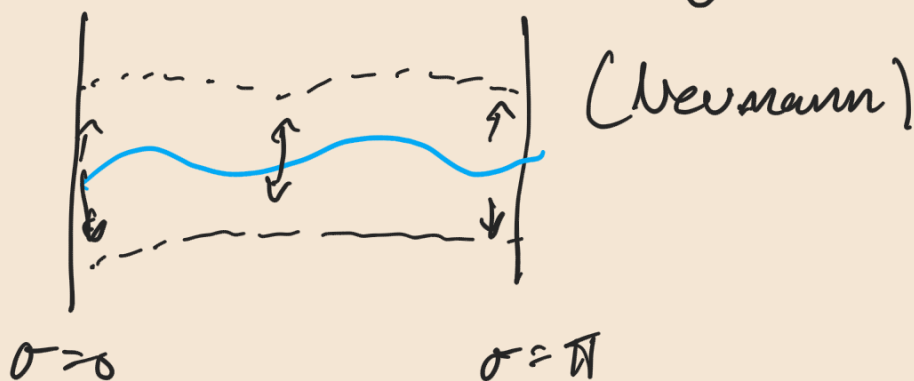
$$T_{01} = T_{10} = \dot{x}^\mu x^{\nu'} \eta_{\mu\nu}$$

B.C's for the open string:

(i) Open w/ Neumann: Set derivative of x^μ by σ to vanish

$$\partial_\sigma x^\mu(\tau, \sigma=0) = \partial_\sigma x^\mu(\tau, \sigma=\pi) = 0$$

→ No momentum is flowing through the ends of the string



• Neumann B.C's preserve Poincaré invariance

$$\begin{aligned} \text{Proof } \partial_\sigma (x^{\mu'})|_{\sigma=0,\pi} &= \partial_\sigma (a_{\nu'}^\mu x^\nu + a^{\mu'})|_{\sigma=0,\pi} \\ &= a_{\nu'}^\mu (\partial_\sigma x^\nu)|_{\sigma=0,\pi} = 0 \quad \checkmark \end{aligned}$$

(ii) Open w/ Dirichlet:

$$x^\mu(\tau, \sigma=0) \equiv x_0^\mu \in \mathbb{R} \quad (\text{constant})$$

$$x^\mu(\tau, \sigma=\pi) \equiv x_\pi^\mu \in \mathbb{R} \quad (\text{constant})$$

• Dirichlet B.C's Do NOT Preserve Poincaré invariance

$$\begin{aligned} \text{Proof: } x^{\mu'}|_{\sigma=0,\pi} &= (a_{\nu'}^\mu x^\nu + a^{\mu'})|_{\sigma=0,\pi} \\ &= a_{\nu'}^\mu x_{0,\pi}^\nu + a^{\mu'} \\ &\neq x_{0,\pi}^{\mu'} \end{aligned}$$

* The conjugate momentum $P^\mu = \frac{\partial \mathcal{L}}{\partial (\partial_\tau X^\mu)}$

$$P^\mu = \dot{X}^\mu / 2\pi\alpha'$$

$$\{X^\mu(\sigma), X^\nu(\sigma')\}_{P.B.} = 0 = \{P^\mu(\sigma), P^\nu(\sigma')\}_{P.B.}$$

$$\{X^\mu(\sigma), P^\nu(\sigma')\}_{P.B.} = \eta^{\mu\nu} \delta(\sigma - \sigma')$$

(c) Closed String: $(\tau, \sigma), 0 \leq \sigma \leq 2\pi$

E.O.M $\partial^2 X^\mu = 0$ subject to B.C.'s

$$X^\mu(\tau, \sigma) = X^\mu(\tau, \sigma + 2\pi)$$

$$(-\partial_\tau^2 + \partial_\sigma^2) X^\mu(\tau, \sigma) = 0$$

We decompose $X^\mu(\tau, \sigma) = A(\sigma) T^\mu(\tau)$

Q: How do I know where μ goes? on A or T ? $\rightarrow$ I don't think it matters...

$$(-\partial_\tau^2 + \partial_\sigma^2)(A(\sigma) T^\mu(\tau))$$

$$(\partial_\sigma^2 A) T^\mu(\tau) - A(\sigma) \partial_\tau^2 T^\mu(\tau) = 0$$

$$\partial_\sigma^2 A(\sigma) = -\lambda^2 A(\sigma)$$

$$A(\sigma) = \tilde{A} e^{i\lambda\sigma} + \tilde{B} e^{-i\lambda\sigma}$$

$$A(\sigma) = \tilde{A} + \tilde{B} = \tilde{A} e^{i\lambda 2\pi} + \tilde{B} e^{-i\lambda\sigma}$$

$$\cos(\lambda 2\pi) = 1$$

$$\lambda 2\pi = n 2\pi$$

$$\therefore \lambda = n \in \mathbb{Z}$$

$$A_n(\sigma) = \tilde{A}_n e^{in\sigma}, \quad \forall n \in \mathbb{Z}$$

If we then sum over all possible states

$$X^\mu(\sigma, \tau) = \sum_{n=-\infty}^{\infty} e^{in\sigma} x_n^\mu(\tau)$$

→ To ensure $x^\mu(\sigma, \tau)$ is real we have

$$(x_n^\mu(\tau))^* = (x_{-n}^\mu(\tau))$$

→ Plugging this into $(-\partial_\tau^2 + \partial_\sigma^2)x^\mu(\sigma, \tau) = 0$

$$(-\partial_\tau^2 + \partial_\sigma^2) \sum_{n=-\infty}^{\infty} e^{in\sigma} x_n^\mu(\tau) = 0$$

$$\Rightarrow \sum_{n=-\infty}^{\infty} \left(-\ddot{x}_n^\mu(\tau) e^{in\sigma} + (in)^2 e^{in\sigma} x_n^\mu(\tau) \right) \\ = \sum_{n=-\infty}^{\infty} e^{in\sigma} \left[\ddot{x}_n^\mu(\tau) + n^2 x_n^\mu(\tau) \right] = 0$$

or $\ddot{x}_n^\mu(\tau) = -n^2 x_n^\mu(\tau)$

(As expected...)

• The lowest mode $n=0$:

$$\ddot{x}_0^\mu(\tau) = 0$$

Is the Center of Mass coordinate

$$x_0^\mu(\tau) = \tilde{x}^\mu + p^\mu \tau$$

* To get to the Point of Quantization we introduce the Creation & ann. operators (Much like the Harmonic oscillator):

$$\gamma^\mu(\tau, \sigma) \equiv \frac{1}{\sqrt{2\alpha'}} \left(\tilde{x}^\mu(\tau, \sigma) + x^\mu(\tau, \sigma) \right)$$

$$\tilde{\gamma}^\mu(\tau, \sigma) \equiv \frac{1}{\sqrt{2\alpha'}} \left(\tilde{x}^\mu(\tau, \sigma) - x^\mu(\tau, \sigma) \right)$$

(Compare to $a = \frac{1}{\sqrt{2}}(\hat{x} + i\hat{p})$, $a^\dagger = \frac{1}{\sqrt{2}}(\hat{x} - i\hat{p})$)

* To solve for $\gamma^\mu(\tau, \sigma)$, note

$$\gamma^\mu(\tau, \sigma) = \frac{1}{\sqrt{2\alpha'}} (\partial_\tau + \partial_\sigma) x^\mu(\tau, \sigma)$$

Thus, we have the first order

$$(\partial_\sigma - \partial_\tau) \gamma^M(\tau, \sigma) = 0$$

$$\gamma^M(\tau, \sigma) = \alpha^M(\tau) A(\sigma)$$

$$\text{Subject to } A(0) = A(2\pi)$$

$$A'(\sigma) \alpha^M(\tau) = \dot{\alpha}^M(\tau) A(\sigma)$$

$$A(\sigma) = \tilde{A} e^{i\lambda\sigma}, \quad \lambda = n \in \mathbb{Z}$$

as before, thus

$$\dot{\alpha}_n^M(\tau) = -n \alpha_n^M(\tau)$$

Summing over all possible states

$$\gamma^M(\tau, \sigma) = \sum_{n=-\infty}^{\infty} \alpha_n^M(\tau) e^{-in\sigma}$$

where all constants are absorbed in α_n^M

$$\gamma^M(\tau, \sigma) = \sum_{n=-\infty}^{\infty} \alpha_n^M(\tau) e^{-in\sigma}$$

$$\bar{\gamma}^M(\tau, \sigma) = \sum_{n=-\infty}^{\infty} \bar{\alpha}_n^M(\tau) e^{in\sigma}$$

* With the conjugate momentum, we can see that $P^M = \dot{X}^M / 2\pi\alpha'$

[Q: It is evident that $T_{0M} \neq P^M$, why is this?

• The total momentum is given by

$$P^M = \int_{-\pi}^{\pi} d\sigma P^M = \int_{-\pi}^{\pi} d\sigma \dot{X}^M / 2\pi\alpha'$$

$$= \int_{-\pi}^{\pi} d\sigma \sum_{n=-\infty}^{\infty} e^{in\sigma} \dot{X}_n^M(\tau) / 2\pi\alpha'$$

• Assuming absolute convergence... $\sum \int \leftrightarrow \int \sum$

$$\sum_{n=-\infty}^{\infty} \left(\int_{-\pi}^{\pi} d\sigma e^{in\sigma} \right) \dot{X}_n^M(\tau) = \sum_{n \neq 0} \left(\frac{1}{in} e^{in\sigma} \right) \dot{X}_n^M(\tau) \Big|_{-\pi}^{\pi}$$

$$= 2\pi \dot{X}_0^M(\tau) / (2\pi\alpha')$$

$$e^{in\pi} = \cos(n\pi) + i\sin(n\pi) \quad n \neq 0$$

$$n=0 \Rightarrow 1$$

$$n \geq 1 \Rightarrow -1 = (-1)^n$$

$$n \geq 2$$

$$e^{-in\pi} = 1 \quad n=0$$

$$n=1 \Rightarrow -1$$

$$\therefore \left(\sum_{n \neq 0} \frac{1}{in} (-1)^n \dot{X}_n^\mu \right) - \left(\sum_{n \neq 0} \frac{1}{in} (-1)^n \dot{X}_n^\mu \right) = 0$$

Leaving

$$p^\mu = \frac{\dot{X}_0^\mu(\tau)}{\alpha} \quad (\text{total momentum})$$

* Claim:

$$p^\mu = \frac{\dot{X}_0^\mu}{\alpha} = \frac{1}{\sqrt{2\alpha'}\pi} \int_{-\pi}^{\pi} d\sigma p^\mu = \frac{1}{\sqrt{2\alpha'}\pi} \int_{-\pi}^{\pi} d\sigma \mathcal{P}^\mu = \sqrt{\frac{2}{\alpha'}} \alpha_0^\mu = \sqrt{\frac{2}{\alpha'}} \bar{\alpha}_0^\mu$$

$$\text{where } \alpha_0^\mu = \bar{\alpha}_0^\mu$$

Proof:

$$\frac{1}{\sqrt{2\alpha'}\pi} \int_{-\pi}^{\pi} d\sigma \left(\frac{1}{\sqrt{2\alpha'}} (\dot{X}^\mu(\tau, \sigma) + X'^\mu(\tau, \sigma)) \right)$$

$$\text{Using } X^\mu(\tau, \sigma) = \sum_{n=-\infty}^{\infty} e^{in\sigma} X_n^\mu(\tau)$$

$$\dot{X}^\mu = \sum e^{in\sigma} \dot{X}_n^\mu(\tau), \quad X'^\mu = \sum in e^{in\sigma} X_n^\mu(\tau)$$

$$\frac{1}{\sqrt{2\alpha'}\pi} \int_{-\pi}^{\pi} d\sigma \sum_{n=-\infty}^{\infty} (e^{in\sigma}) (\dot{X}_n^\mu(\tau) + in X_n^\mu(\tau))$$

$$= 0 \quad \forall n \neq 0$$

$$n=0: \frac{1}{\sqrt{2\alpha'}\pi} \int_{-\pi}^{\pi} d\sigma \dot{X}_0^\mu = \frac{\dot{X}_0^\mu}{\alpha} \quad \checkmark$$

In terms of raising & lowering ops

$$p^\mu = \frac{1}{\sqrt{2\alpha'}\pi} \int_{-\pi}^{\pi} d\sigma \sum_{n=-\infty}^{\infty} \alpha_n^\mu(\tau) e^{in\sigma} = \frac{2\pi}{\sqrt{2\alpha'}\pi} \alpha_0^\mu(\tau)$$

$$p^\mu = \dot{X}_0^\mu / \alpha' = \sqrt{\frac{2}{\alpha'}} \alpha_0^\mu(\tau)$$

* Since $\mathcal{P}^\mu = \frac{1}{\sqrt{2\alpha'}} (\dot{X}^\mu + X'^\mu)$ and $p^\mu = \frac{\dot{X}^\mu}{2\pi\alpha'}$

$$\mathcal{P}^\mu = \frac{1}{\sqrt{2\alpha'}} (2\pi\alpha' p^\mu + X'^\mu)$$

$$= \sqrt{2\pi\alpha'} p^\mu + \frac{1}{\sqrt{2\alpha'}} X'^\mu$$

$$\bar{p}^M = \sqrt{2\alpha'} \pi P^M - \frac{X'^M}{\sqrt{2\alpha'}}$$

with

$$\{x^\mu, x^\nu\} = 0 \Rightarrow \{p^\mu, p^\nu\}$$

$$\{x^\mu, p^\nu\} = \eta^{\mu\nu} \delta(\sigma - \sigma')$$

$$\Rightarrow \{\bar{p}^M(\sigma), \bar{p}^N(\sigma')\} = \left\{ \sqrt{2\alpha'} \pi P^M + \frac{1}{\sqrt{2\alpha'}} X'^M, \sqrt{2\alpha'} \pi P^N - \frac{1}{\sqrt{2\alpha'}} X'^N \right\}$$

$$= \{P^M, P^N\} - \{P, X\} + \{X, P^N\} - \{X, X\}$$

$$= 0$$

$$\therefore \{\bar{p}^M(\sigma), \bar{p}^N(\sigma')\}_{PB} = 0$$

$$\bullet \{p^\mu(\sigma), p^\nu(\sigma')\} = \left\{ \sqrt{2\alpha'} \pi P^\mu + \frac{1}{\sqrt{2\alpha'}} X'^\mu, \sqrt{2\alpha'} \pi P^\nu - \frac{1}{\sqrt{2\alpha'}} X'^\nu \right\}$$

$$= \{P, P\} + \{X', X'\} + 2 \{X'^\mu, P^\nu\}$$

$$= 2 \left\{ \frac{1}{\sqrt{2\alpha'}} \frac{d}{d\sigma} X^\mu, \sqrt{2\alpha'} \pi P^\nu \right\}$$

$$= 2\pi \frac{d}{d\sigma} (\eta^{\mu\nu} \delta(\sigma - \sigma'))$$

$$\therefore \{p^\mu(\sigma), p^\nu(\sigma')\} = 2\pi \eta^{\mu\nu} \frac{d}{d\sigma} (\delta(\sigma - \sigma'))$$

* Similarly we find that

$$\{\bar{p}^\mu(\sigma), \bar{p}^\nu(\sigma')\} = -2\pi \eta^{\mu\nu} \frac{d}{d\sigma} (\delta(\sigma - \sigma'))$$

• We can now find commutation relations for all of the modes α_n^μ :

$$p^\mu(\sigma, \tau) = \sum_{n=-\infty}^{\infty} \alpha_n^\mu(\tau) e^{in\sigma}$$

$$\{p^\mu, p^\nu\} = \sum_{n,m} e^{-in\sigma} e^{-im\sigma'} \{\alpha_n^\mu(\tau), \alpha_m^\nu(\tau)\}$$

Now, Recall that the delta fn has

the expansion

$$\delta(\sigma - \sigma') = \frac{1}{2\pi} \sum_{n=-\infty}^{\infty} e^{in(\sigma - \sigma')}$$

$$\sum_{n,m} e^{-i(n\sigma+m\sigma')} \{ \alpha_n^u(\tau), \alpha_m^v(\tau) \} = \frac{2\pi}{2\pi} \eta^{\mu\nu} \frac{d}{d\sigma} \sum_{n=-\infty}^{\infty} e^{2in(\sigma-\sigma')}$$

$$\sum_{n,m} e^{-i(n\sigma+m\sigma')} \{ \alpha_n^u(\tau), \alpha_m^v(\tau) \} = \eta^{\mu\nu} \sum_{n=-\infty}^{\infty} \frac{1}{in} e^{2in(\sigma-\sigma')}$$

RHS:

$$i\eta^{\mu\nu} \sum_{n=-\infty}^{\infty} n e^{i(n\sigma - n\sigma')}$$

$$= i\eta^{\mu\nu} \sum_{m,n} (\delta_{n+m,0}) n e^{i(n\sigma + m\sigma')}$$

Since $\delta(\sigma' - \sigma) = \delta(\sigma - \sigma')$ $e^i \rightarrow e^{-i}$

$$n \rightarrow -n, m \rightarrow -m$$

$\therefore$ Thus:

$$\{ \alpha_n^u, \alpha_m^v \} = i\eta^{\mu\nu} (-n) \delta_{n+m,0}$$

Similarly for $\bar{\alpha}^u$:

*We obtain

$$\{ \alpha_n^u, \alpha_m^v \} = \{ \bar{\alpha}_n^u, \alpha_m^v \} = -in\eta^{\mu\nu} \delta_{n+m,0}$$

$$\{ \alpha_n^u, \bar{\alpha}_m^v \} = 0$$

Now using $x_0^u, \alpha_0^u, \alpha_n^u(\tau), \bar{\alpha}_n^u(\tau), n \neq 0$
to describe the phase space (p_n, x_n)

• Now, we obtain the E.O.M for $\alpha_n^u, \dots$

Recall $\ddot{X}^\mu = \ddot{X}''^\mu$

$$\frac{\partial}{\partial \tau} p^\mu = \sum_n \dot{\alpha}_n^\mu e^{-in\sigma}$$

$$\frac{\partial}{\partial \sigma} p^\mu = \frac{1}{\sqrt{2\alpha'}} (\ddot{X}^\mu + \dot{X}'^\mu)$$

$$\frac{\partial}{\partial \sigma} p^\mu = \frac{1}{\sqrt{2\alpha'}} (\ddot{X}'^\mu + \ddot{X}''^\mu)$$

But $\ddot{X}^\mu = \ddot{X}''^\mu$ so we find that

$$\partial_\tau p^\mu = \partial_\sigma p^\mu$$

$$\sum_n \dot{\alpha}_n^\mu e^{-in\sigma} = \sum_n \alpha_n^\mu (-in) e^{-in\sigma}$$

$$\dot{\alpha}^\mu(\tau) = -in \alpha^\mu(\tau)$$

$$\alpha_n^\mu(\tau) = \alpha_n^\mu(0) e^{-in\tau}$$

$$\bar{\alpha}_n^\mu(\tau) = \bar{\alpha}_n^\mu(0) e^{-in\tau}$$

Recall that $\frac{\dot{X}_0^\mu}{\alpha'} = \sqrt{\frac{2}{\alpha'}} \alpha_0^\mu(\tau)$

where $\alpha_0^\mu(\tau) =: \alpha_0^\mu(0)$

$$\Rightarrow \dot{X}_0^\mu(\tau) = \sqrt{2\alpha'} \alpha_0^\mu$$

$$X_0^\mu(\tau) = \sqrt{2\alpha'} \alpha_0^\mu \tau + q^\mu$$

Here $q^\mu = X_0^\mu(0)$

* Since

$$\mathcal{P}^\mu = \frac{1}{\sqrt{2\alpha'}} (\dot{X}^\mu + \dot{X}^{\mu})$$

$$\bar{\mathcal{P}}^\mu = \frac{1}{\sqrt{2\alpha'}} (\dot{X}^\mu - \dot{X}^{\mu})$$

$$\Rightarrow \sqrt{2\alpha'} (\mathcal{P}^\mu - \bar{\mathcal{P}}^\mu) = 2\dot{X}^{\mu}$$

$$(\dot{X}^{\mu})^\mu = \sqrt{\frac{\alpha'}{2}} (\mathcal{P}^\mu - \bar{\mathcal{P}}^\mu)$$

$$= \sqrt{\frac{\alpha'}{2}} \left[\sum_{n=-\infty}^{\infty} (\alpha_n^\mu(0) e^{-in\tau} e^{-in\sigma} - \bar{\alpha}_n^\mu(0) e^{in\tau} e^{in\sigma}) \right]$$

$$\frac{d}{d\sigma} X^\mu = \text{RHS}$$

• If we now integrate this over the closed string:

$$\int_{-\pi}^{\pi} d\sigma \frac{d}{d\sigma} X^\mu(\tau, \sigma) = X^\mu(\tau, \sigma) \Big|_{\sigma=0}^{\sigma=2\pi}$$

$$+ \sqrt{\frac{\alpha'}{2}} \sum_{n=-\infty}^{\infty} \left[\alpha_n^\mu(0) e^{-in\tau} (e^{-in\sigma} / -in) - \bar{\alpha}_n^\mu(0) e^{-in\tau} (e^{in\sigma} / in) \right] \Big|_{\sigma=0}^{\sigma=2\pi}$$

Recall $X^\mu(\tau, \sigma) = \sum_{n=-\infty}^{\infty} \alpha_n^\mu(\tau) e^{-in\sigma}$

$$\Rightarrow X^\mu(\tau, 0) = \sum_{n=-\infty}^{\infty} \alpha_n^\mu(\tau)$$

$\Rightarrow$ NO, instead we isolate $n=0$

$$\sqrt{\frac{\alpha'}{2}} (\alpha_0^\mu - \bar{\alpha}_0^\mu) = 0 \quad \text{since } \alpha_0^\mu = \bar{\alpha}_0^\mu$$

$$X(\tau, \sigma) = x_0^\mu(\tau) + i\sqrt{\frac{\alpha'}{2}} \sum_{n \neq 0} \frac{2}{n} \left[\alpha_n^\mu(\omega) e^{-in(\tau+\sigma)} + \bar{\alpha}_n^\mu(\omega) e^{-in(\tau-\sigma)} \right]$$

$$\text{Since } X(\tau, \sigma) = \sum_{n=-\infty}^{\infty} = \underbrace{X_0^\mu(\tau)}_{n=0} + \sum_{n \neq 0}$$

$$x_0^\mu(\tau) = q^\mu + \sqrt{2\alpha'} \alpha_0^\mu \tau$$

$$\text{Since } \sqrt{\frac{\alpha'}{2}} p^\mu = \alpha_0^\mu \Rightarrow \alpha' p^\mu = \sqrt{2\alpha'} \alpha_0^\mu$$

$$X(\tau, \sigma) = q^\mu + \alpha' \tau p^\mu + i\sqrt{\frac{\alpha'}{2}} \sum_{n \neq 0} \frac{2}{n} \left\{ \alpha_n^\mu e^{-in(\tau+\sigma)} + \bar{\alpha}_n^\mu e^{-in(\tau-\sigma)} \right\}$$

This can be written as

$$X(\tau, \sigma) = X^L(\tau+\sigma) + X^R(\tau-\sigma)$$

$$X^L(\tau+\sigma) = q^\mu + \alpha' \frac{p^\mu}{2} (\tau+\sigma) + i\sqrt{\frac{\alpha'}{2}} \sum_{n \neq 0} \frac{\alpha_n^\mu}{n} e^{-in(\tau+\sigma)}$$

" $\tau \leftrightarrow -\sigma$ "

We have only imposed the constraint

$$\partial_n(\sqrt{-g}) g^{\alpha\beta} \partial_\beta x^\mu = 0$$

which was from the x^μ E.O.M. The

$g_{\alpha\beta}$ E.O.M yields

$$T_{\alpha\beta} = 0$$

$$T_{00} = T_{11} = \frac{1}{2} (\dot{x}^\mu \dot{x}^\nu + x'^\mu x'^\nu) \eta_{\mu\nu}$$

$$T_{01} = T_{10} = \dot{x}^\mu x'^\nu \eta_{\mu\nu}$$

Since

$$p^\mu = \frac{1}{\sqrt{2\alpha'}} (\dot{x}^\mu + x'^\mu)$$

$$\begin{aligned} \therefore p^\mu p_\mu &= \frac{1}{2\alpha'} \eta_{\mu\nu} (\dot{x}^\mu + x'^\mu) (\dot{x}^\nu + x'^\nu) \\ &= \frac{1}{2\alpha'} \eta_{\mu\nu} (\dot{x}^\mu \dot{x}^\nu + 2\dot{x}^\mu x'^\nu + x'^\mu x'^\nu) \end{aligned}$$

$$\alpha' p^\mu p_\mu = (T_{00} + T_{11})$$

$$\alpha' T_{\mu\nu} = (100 + 101)$$

or

$$\begin{aligned} p_{\mu} p_{\nu} &= \frac{1}{\alpha'} (T_{00} + T_{01}) \equiv \frac{2}{\alpha'} T_{++} \\ \bar{p}_{\mu} \bar{p}_{\nu} &= \frac{1}{\alpha'} (T_{00} - T_{01}) \equiv \frac{2}{\alpha'} T_{--} \end{aligned}$$

- Since $T_{\alpha\beta}$, as we shall see, breaks up into holomorphic & Anti holomorphic terms $T_{zz} \equiv T(z)$ & $T_{\bar{z}\bar{z}} \equiv \bar{T}(\bar{z})$ with a mode expansion

$$T(z) = \sum_{n=-\infty}^{\infty} L_n z^{-h_T-n}$$

$$h_T (\text{Conformal weight}) \quad h_T = 2 \text{ (2,0 tensor)}$$

Requiring $T=0$ would implicitly imply that $L_n = \bar{L}_n = 0$

- In particular,

$$\begin{aligned} L_n &= \oint \frac{dz}{2\pi i} z^{n+h_T-1} T(z) \\ &= \oint \frac{dz}{2\pi i} z^{n+1} T(z) \end{aligned}$$

so the relationship between $T(z)$ and $T_{++}(\sigma)$ is given by $z = e^{i(\tau+\sigma)}$?

$$T(z) \sim \frac{1}{\alpha'} T_{++}$$

$$\begin{aligned} L_n &= \frac{1}{2\pi\alpha'} \int_{-\pi}^{\pi} d\sigma e^{in\sigma} T_{++} = \frac{1}{4\pi} \int_{-\pi}^{\pi} d\sigma e^{in\sigma} \dot{x}^2 \\ &= \frac{1}{2} \sum_{m=-\infty}^{\infty} \alpha_m^\mu \alpha_{n-m}^\nu \eta_{\mu\nu} \end{aligned}$$

and likewise for $\bar{L}_n$.

$$\begin{aligned} \{L_n, L_m\} &= -i(n-m)L_{n+m} & \{L_n, \bar{L}_m\} &= 0 \\ \{\bar{L}_n, \bar{L}_m\} &= -i(n-m)\bar{L}_{n+m} & & \forall n, m \end{aligned}$$

The Open String: $-\pi \leq \sigma \leq \pi$

* Must Account for Boundary Conditions.

$$X^\mu(\tau, \sigma) = \begin{cases} X^\mu(\tau, \sigma), & 0 \leq \sigma \leq \pi \\ X^\mu(\tau, -\sigma), & -\pi \leq \sigma \leq 0 \end{cases}$$

- Note we have a reflection about $\sigma = 0$.

- Consider $\sigma \rightarrow \sigma \in (-\infty, \infty)$

s.t. $X(\tau, \sigma) = X(\tau, \sigma + 2\pi)$. This is achieved via the fact

$$\begin{aligned} X^\mu(\tau, \pi) &= X^\mu(\tau, -\pi) \\ \Rightarrow \frac{d}{d\sigma}(X^\mu(\tau, \pi)) &= 0 \end{aligned}$$

So $X^\mu(\tau, \sigma)$ is continuous & possess a first derivative at $\sigma = \pm\pi$.

- We can capture the constraints on T_{ab} T_{00}, T_{11}, T_{01} via the Equations:

$$(\partial^\mu \rho_\mu)(\tau, \sigma) = 0 \quad -\pi < \sigma < \pi$$

Requiring that each vanish separately.

$$\rho^\mu(\tau, \sigma) = \frac{1}{2\alpha'} (\dot{X}^\mu(\tau, \sigma) + X'^\mu(\tau, \sigma)), \quad 0 \leq \sigma \leq \pi$$

* To find the extension of $\rho^\mu(\tau, \sigma)$ we simply apply:

$$\rho^\mu(\tau, \sigma) = \begin{cases} \rho^\mu(\tau, \sigma), & 0 \leq \sigma \leq \pi \\ \rho^\mu(\tau, -\sigma), & -\pi \leq \sigma \leq 0 \end{cases}$$

Q: Is this right? Let's find out:

$$\frac{d}{d\tau} X^\mu(\tau, \sigma) = \begin{cases} \dot{X}^\mu(\tau, \sigma) \\ \dot{X}^\mu(\tau, -\sigma) \end{cases}$$

$\dot{X}^\mu$ is an EVEN fⁿ under $\sigma \rightarrow -\sigma$

* However, $\frac{d}{d\sigma} X^\mu(\tau, \sigma) = \begin{cases} \dot{X}^\mu(\tau, \sigma), & 0 \leq \sigma \leq \pi \\ -\dot{X}^\mu(\tau, -\sigma), & -\pi \leq \sigma \leq 0 \end{cases}$

Under $\sigma \rightarrow -\sigma$:

$$\dot{X}^\mu(\tau, -\sigma) = \begin{cases} -\dot{X}^\mu(\tau, \sigma), & 0 \leq \sigma \leq \pi \\ \dot{X}^\mu(\tau, -\sigma), & -\pi \leq \sigma \leq 0 \end{cases}$$

So $\dot{X}^\mu$ is ODD

Boundary Conditions:

$$\frac{d}{d\sigma} X^\mu(\tau, \sigma) = 0, \quad \sigma = 0, \sigma = \pi$$

Recall we had,

$$X^\mu(\tau, \sigma) = T^\mu(\tau) \cdot A(\sigma)$$

$$\Rightarrow \frac{d}{d\sigma} X^\mu = T^\mu(\tau) A'(\sigma) = 0 \quad (\sigma = 0, \pi)$$

It is therefore obvious to have

$$A'(\sigma) = \sin(n\sigma)$$

* The full expansion is therefore

$$X^\mu(\tau, \sigma) = \sum_{n=0}^{\infty} X_n^\mu(\tau) \cos(n\sigma) = X_0^\mu(\tau) + \sum$$

note

$$\sum_{n=0}^{\infty} X_n^\mu(\tau) \cos(n\sigma) = \sum_{n=0}^{\infty} X_n^\mu \frac{1}{2} (e^{in\sigma} + e^{-in\sigma})$$

$$= \frac{1}{2} \sum_{n=0}^{\infty} X_n^\mu(\tau) e^{in\sigma} + \frac{1}{2} \sum_{n=0}^{\infty} X_n^\mu(\tau) e^{-in\sigma}$$

$$= () + \frac{1}{2} (n \rightarrow -n)$$

$$= () + \frac{1}{2} \sum_{n=-\infty}^{\infty} X_{-n}^\mu(\tau) e^{in\sigma}$$

* Note: X^μ is even under $\sigma \rightarrow -\sigma$

as such

$$X_{-n}^\mu(\tau) = X_n^\mu(\tau)$$

$$\underline{X^M}(\tau, \sigma) = X_0^M(\tau) + \sum_{n \neq 0} X_n^M(\tau) e^{in\sigma}$$

* We use orthogonal f_n^s to solve for the coefficients of $\sum : X_n^M(\tau)$

$$\int_{-\pi}^{\pi} d\sigma \underline{X^M}(\tau, \sigma) \cos(m\sigma) = \sum_{n=-\infty}^{\infty} X_n^M(\tau) \int_{-\pi}^{\pi} \cos(n\sigma) \cos(m\sigma) d\sigma$$

$$\underline{n=0} : \int_{-\pi}^{\pi} d\sigma \underline{X^M}(\tau, \sigma) \cos(m\sigma) = \sum_{n=-\infty}^{\infty} X_n^M(\tau) \int_{-\pi}^{\pi} \cos(n\sigma) \cos(m\sigma) d\sigma$$

$$\int_{-\pi}^{\pi} d\sigma \cos(n\sigma) \cos(m\sigma) = \begin{cases} 0 & n \neq m \\ \pi & n = m \neq 0 \\ 2\pi & n = m = 0 \end{cases}$$

* $n=0$, then $m=0$

$$X_0^M(\tau) 2\pi = \int_{-\pi}^{\pi} d\sigma \underline{X^M}(\tau, \sigma)$$

$$\text{or } X_0^M(\tau) = \left(\frac{1}{2\pi}\right) \int_{-\pi}^{\pi} d\sigma \underline{X^M}(\tau, \sigma)$$

$n=m \neq 0$

$$X_n^M(\tau) = \frac{2}{\pi} \int_0^{\pi} d\sigma' \cos(n\sigma') \underline{X^M}(\tau, \sigma'), \quad n \neq 0$$

* The same Equation of motion holds for the open string, $\partial^2 \underline{X^M} = 0$

$$(-\partial_{\tau}^2 + \partial_{\sigma}^2) \underline{X^M}(\tau, \sigma)$$

$$(\partial_{\tau}^2 + \partial_{\sigma}^2) \left\{ \sum_{n=-\infty}^{\infty} X_n^M(\tau) \cos(n\sigma) \right\}$$

$$= \sum_{n=-\infty}^{\infty} \left\{ (-\ddot{X}_n^M(\tau) - n^2 X_n^M(\tau)) \cos(n\sigma) \right\}$$

We could argue that at each "level" the above $\{ \}$ must vanish but a

better way to see this is to integrate using orthogonal fns. [TRIVIAL]:

$$\therefore \boxed{\ddot{X}_n^{\mu}(\tau) + n^2 X_n^{\mu}(\tau) = 0}$$

* $n \geq 1$ we have an infinite # of harmonic oscillators plus a center of mass coordinate: $X_0^{\mu}(\tau) = q^{\mu} + p^{\mu} \tau$

→ Still need to introduce the constraint.

$$\tau_{03} = 0$$

$$\boxed{p^{\mu}(\tau, \sigma) \equiv \sum_{-\infty}^{\infty} \alpha_n^{\mu}(\tau) e^{in\sigma}}$$

using ortho-fns

$$\boxed{\alpha_n^{\mu}(\tau) = \frac{1}{2\pi} \int_{-\pi}^{\pi} d\sigma e^{in\sigma} p^{\mu}(\tau, \sigma)}$$

* Using $\tilde{X}(-\sigma) = \tilde{X}(\sigma)$ and $\tilde{X}'(-\sigma) = -X'(\sigma)$

$$p^{\mu}(\tau, \sigma) - p^{\mu}(\tau, -\sigma) = \frac{2X'^{\mu}(\tau, \sigma)}{\sqrt{2\alpha'}} \quad \text{[Equation 2.2.10]}$$

$$\int_{-\pi}^{\pi} d\sigma \frac{2X'^{\mu}(\tau, \sigma)}{\sqrt{2\alpha'}} = \int_{-\pi}^{\pi} d\sigma \{p^{\mu}(\tau, \sigma) - p^{\mu}(\tau, -\sigma)\}$$

$$\begin{aligned} X^{\mu}(\tau, \sigma) &= \int_{-\pi}^{\pi} d\sigma \sum_{n=-\infty}^{\infty} \left\{ \alpha_n^{\mu}(\tau) e^{-in\sigma} - \alpha_{-n}^{\mu}(\tau) e^{in\sigma} \right\} \\ &= \alpha_0^{\mu}(\tau) + \frac{2\pi}{\sqrt{2\alpha'}} (\alpha_0^{\mu}(\tau) - \alpha_0^{\mu}(\tau)) \\ &\quad + i\sqrt{\frac{\alpha'}{2}} \sum_{n \neq 0} \frac{2}{n} (\alpha_n^{\mu}(\tau) - \alpha_{-n}^{\mu}(\tau)) e^{-in\sigma} \end{aligned}$$

$$\therefore \underline{\tilde{X}^{\mu}}(\tau, \sigma) = X_0^{\mu}(\tau) + i\sqrt{\frac{\alpha'}{2}} \sum_{\substack{n \neq 0 \\ -\infty}}^{\infty} \frac{2}{n} (\alpha_n^{\mu}(\tau) - \alpha_{-n}^{\mu}(\tau)) e^{-in\sigma}$$

E.O.M: $(-\partial_{\tau}^2 + \partial_{\sigma}^2) \tilde{X}^{\mu}(\tau, \sigma) = 0$

→ $X_0^{\mu}(\tau)$ satisfies this separately so that

$$\dot{\alpha}_n^\mu(\tau) = -in\alpha_n^\mu(\tau) \Rightarrow \alpha_n^\mu(\tau) = e^{-in\tau} \alpha_n^\mu(0)$$

Note, at the level of operators this expression is given by

$$\mathcal{P}^\mu = \dot{X}^\mu \quad \text{or}$$

* The zero mode term:

$$\dot{X}^\mu / \sqrt{2\alpha'} = (\mathcal{P}^\mu(\tau, \sigma) + \mathcal{P}^\mu(\tau, -\sigma))$$

$$2\dot{X}_0^\mu / \sqrt{2\alpha'} = \frac{2}{2\pi} \int_{-\pi}^{\pi} d\sigma (\mathcal{P}^\mu(\tau, \sigma) + \mathcal{P}^\mu(\tau, -\sigma)) = 2\alpha'_0$$

$$\dot{X}_0^\mu(\tau) = \sqrt{2\alpha'} \alpha_0^\mu$$

$$\therefore q^\mu \equiv X_0^\mu(0)$$

$$X_0^\mu(\tau) = q^\mu + \sqrt{2\alpha'} \alpha_0^\mu \tau$$

$$X^\mu(\tau, \sigma) = q^\mu + \sqrt{2\alpha'} \alpha_0^\mu \tau + i\sqrt{\frac{\alpha'}{2}} \sum_{n \neq 0} \frac{1}{n} (\alpha_n^\mu(0) e^{-in(\tau+\sigma)} + \alpha_{-n}^\mu(0) e^{-in(\tau-\sigma)})$$



Open-string Expansion



Using the relation $\mathcal{P}^\mu(0) = \sqrt{2\alpha'} \pi \dot{X}^\mu + \frac{X'^\mu}{\sqrt{2\alpha'}}$ we find

$$\{\mathcal{P}^\mu(\sigma), \mathcal{P}^\nu(\sigma')\} = 2\pi\alpha' \frac{d}{d\sigma} \delta(\sigma - \sigma')$$

* Again, expanding in Fourier comp's

$$\{\alpha_n^\mu, \alpha_m^\nu\} = -in\delta_{n+m,0} \eta^{\mu\nu}$$

• Total momentum: $P^\mu = \int_0^\pi d\sigma \mathcal{P}^\mu = \frac{\dot{X}_0^\mu}{2\alpha'} = \frac{\alpha_0^\mu}{\sqrt{2\alpha'}}$

$$M^2 = -P^\mu P_\mu = -\alpha_0^\mu \alpha_{0\mu} = -\frac{1}{2\alpha'} \alpha_0^\mu \alpha_{0\mu}$$

Note: $\{X^\mu(\sigma), P^\nu(\sigma')\} = \eta^{\mu\nu} \delta(\sigma - \sigma')$

$$\int_0^{2\pi} d\sigma \{X^\mu(\sigma), P^\nu(\sigma')\} = \eta^{\mu\nu}$$

$$\int_0^{2\pi} d\sigma X^\mu(\sigma) = X^\mu_0 \quad \text{and so}$$

$$\{q^\mu, p^\nu\} = \eta^{\mu\nu}$$

E.M. Tensor & Angular Mom. of the string:

* Poincaré Symmetry:

$$-\pi^{\mu\alpha} = - \frac{\partial \mathcal{L}}{\partial (\partial_\alpha X_\mu)} \quad (\text{translational current})$$

$$(M^\alpha)_{\mu\nu} = X^\mu \frac{\partial \mathcal{L}}{\partial (\partial_\alpha X_\nu)} - (\mu \leftrightarrow \nu) = X^\mu \pi^{\nu\alpha} - X^\nu \pi^{\mu\alpha}$$

(Lorentz-rotations)

Using the Eul.-Lag. eq's these currents are conserved:

$$\partial_\alpha \pi^\mu_\alpha = 0, \quad \partial_\alpha M^\mu_{\alpha\nu} = 0$$

$$S = \frac{1}{2\pi\alpha'} \int d^2\sigma \sqrt{-g} g^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X^\nu \eta_{\mu\nu}$$

$g_{\alpha\beta} \approx \eta_{\alpha\beta}$, then

$\Leftrightarrow$

$$\pi^\mu_\alpha = \frac{1}{2\pi\alpha'} \partial_\alpha X^\mu$$

likewise

$$(M_\alpha)^{\mu\nu} = \frac{1}{2\pi\alpha'} (X^\mu \partial_\alpha X^\nu - X^\nu \partial_\alpha X^\mu)$$

* Take a closed curve C on the worldsheet with surface S within





Then, using Stokes Thm

$$\int_C (\pi_0^M d\sigma + \pi_1^M d\tau) = - \int_S d\tau d\sigma \partial^\alpha \pi_\alpha^M = 0$$

In terms of differential forms:

$$\hat{T}^M = G_{\alpha\beta} \pi^{\alpha\beta} d\zeta^B$$

Flow of energy/momentum through any open curve C on the worldsheet.
 $\Rightarrow$ flow of E & P through the ends of the string are zero.

A Classical solution of the open string:

$\rightarrow$ Find explicit sol^{ns} for

$$\partial_\alpha \partial^\alpha x^M = 0 \quad \& \quad T_{\alpha\beta} = 0$$

$$\bullet \quad t \equiv x^0 = \tau d, \quad d \in \mathbb{R}$$

$$T_{00} \Leftrightarrow \frac{1}{2} (\dot{x}^M \dot{x}^N + x^{M'} x^{N'}) \eta_{MN} = 0$$

$$- \dot{x}^0 \dot{x}^0 + \sum_{i=1}^{D-1} \dot{x}^i \dot{x}^i - x^{0'} x^{0'} + \dots$$

